

Lagrange Duality: Dual Function Construction, Strong Duality, and Sensitivity Analysis

Numerical Optimization

Interactive implementation on the natID framework

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Abstract. The Lagrange dual function is constructed by pointwise minimisation of the Lagrangian, and weak duality, strong duality, Slater’s condition and the shadow-price reading of the multipliers are derived. Three instances are implemented: a strictly convex quadratic program whose dual is available in closed form, a resource-allocation problem reproducing the water-filling rule, and a non-convex problem where strong duality fails. In the convex cases the primal is solved twice by unrelated methods, so the reported duality gap is a cross-validation rather than one number printed twice. For the non-convex instance the perturbation function and its convex envelope exhibit the gap directly. All results agree with theory to between 10^{-11} and 10^{-16} .

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1 Introduction

Duality attaches to every constrained minimisation problem a second, always-convex problem built from its constraints. The dual's value bounds the primal's from below, under mild conditions equals it, and its solution prices the constraints. This report derives that theory, applies it to three instances, and describes the interactive application built to compute and draw it. Every figure was produced by that application.

2 Mathematical foundations

2.1 The Lagrangian and the dual function

For the problem in standard form

$$\min_{x \in \mathbb{R}^n} f_0(x) \quad \text{subject to} \quad f_i(x) \leq 0, \quad i = 1, \dots, m, \quad (1)$$

with optimal value p^* , the **Lagrangian** attaches a price λ_i to each constraint,

$$L(x, \lambda) = f_0(x) + \sum_{i=1}^m \lambda_i f_i(x), \quad (2)$$

and the **dual function** is its unconstrained infimum over x ,

$$g(\lambda) = \inf_{x \in \mathbb{R}^n} L(x, \lambda), \quad (3)$$

which may equal $-\infty$. The **dual problem** is $\max_{\lambda \geq 0} g(\lambda)$, with optimal value d^* . No convexity is assumed anywhere above.

Proposition 1. For every problem of the form (1) — convex or not — g is concave.

Proof. For fixed x , the map $\lambda \mapsto L(x, \lambda)$ is **affine**, since $f_0(x)$ and $f_i(x)$ are then constants. By (3), g is a pointwise infimum of affine functions, and such an infimum is concave. ■

This is the structural reason duality is useful: whatever pathology the primal has, **the dual is a concave maximisation over a convex set**. Section 5 exhibits a non-convex primal with a perfectly well-behaved concave dual.

2.2 Weak duality

Theorem 1 (Weak duality). For every $\lambda \geq 0$, $g(\lambda) \leq p^*$; hence $d^* \leq p^*$.

Proof. Let $\tilde{x}$ be feasible and $\lambda \geq 0$. Each $\lambda_i f_i(\tilde{x})$ is non-positive, so

$$g(\lambda) = \inf_x L(x, \lambda) \leq L(\tilde{x}, \lambda) = f_0(\tilde{x}) + \sum_i \lambda_i f_i(\tilde{x}) \leq f_0(\tilde{x}). \quad (4)$$

Taking the infimum over feasible $\tilde{x}$ and the supremum over $\lambda \geq 0$ gives the claim. ■

Only the sign conditions $\lambda \geq 0$ and $f_i(\tilde{x}) \leq 0$ were used — **no convexity**. Weak duality is therefore universal, which is what makes dual bounds valuable in non-convex optimisation. The quantity $p^* - d^* \geq 0$ is the **duality gap**.

2.3 Strong duality, Slater's condition, and the KKT system

Equality is not automatic. **Slater's condition** is the standard sufficient criterion: if the problem is convex and there exists a **strictly feasible** point ($f_i(\tilde{x}) < 0$ for all i), then $d^* = p^*$ and the dual optimum is attained. It is sufficient, not necessary.

When strong duality holds and both optima are attained at x^*, λ^* , the chain

$$f_0(x^*) = g(\lambda^*) \leq f_0(x^*) + \sum_i \lambda_i^* f_i(x^*) \leq f_0(x^*) \quad (5)$$

begins and ends at the same value, so both inequalities are equalities. The second forces every term of $\sum_i \lambda_i^* f_i(x^*) -$ each ≤ 0 – to vanish:

$$\lambda_i^* f_i(x^*) = 0, \quad i = 1, \dots, m. \quad (6)$$

This is **complementary slackness**: a constraint is either active and may carry any non-negative price, or slack and priced at zero. The first equality says x^* minimises $L(\cdot, \lambda^*)$ over all of $\mathbb{R}^n$, so in the differentiable case

$$\nabla f_0(x^*) + \sum_i \lambda_i^* \nabla f_i(x^*) = 0 \quad (7)$$

(**stationarity**). With primal and dual feasibility these are the **KKT conditions**, necessary and sufficient for a convex problem satisfying Slater. All four residuals are measured in Section 7.

2.4 Sensitivity: shadow prices and the convex envelope

Perturbing the constraint levels defines the **perturbation function**

$$p(u) = \inf\{f_0(x) : f_i(x) \leq u_i\}, \quad p(0) = p^*. \quad (8)$$

Theorem 2 (Global sensitivity). If strong duality holds with dual optimum λ^* , then $p(u) \geq p^* - (\lambda^*)^T u$ for every u .

Proof. If $f_i(x) \leq u_i$ then, using $\lambda^* \geq 0$, $p^* = g(\lambda^*) \leq f_0(x) + \sum_i \lambda_i^* f_i(x) \leq f_0(x) + (\lambda^*)^T u$. Rearranging and taking the infimum gives the claim. ■

So $u \mapsto p^* - (\lambda^*)^T u$ is a supporting hyperplane touching p at the origin, and $-\lambda^*$ is a subgradient of p there. Where p is differentiable,

$$\frac{\partial p^*}{\partial u_i} = -\lambda_i^* : \quad (9)$$

λ_i^* is the **shadow price** of constraint i , the rate at which the optimal cost falls per unit of relaxation. An inactive constraint has $\lambda_i^* = 0$ by (6) and so prices at zero, as intuition demands.

The same perturbation view explains gaps. One shows $g(\lambda) = \inf_u (p(u) + \lambda^T u)$, whence d^* is the value at $u = 0$ of the largest convex function lying below p – its **convex envelope**:

$$p^* = p(0), \quad d^* = (\text{conv } p)(0), \quad \text{gap} = p(0) - (\text{conv } p)(0). \quad (10)$$

If p is convex near the origin the two coincide. If p has a non-convex dent there, the envelope cuts a chord across it and **the gap is the depth of the dent** – computed directly in Section 5.

3 Instance A: convex quadratic program

3.1 Model and closed-form dual

$$\min_{x \in \mathbb{R}^n} \frac{1}{2} x^T P x + q^T x \quad \text{subject to} \quad A x \leq b, \quad P \succ 0, \quad (11)$$

with default data

$$P = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}, \quad q = \begin{pmatrix} -3 \\ -4 \end{pmatrix}, \quad A = \begin{pmatrix} 1 & 1 \\ -1 & 2 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 4 \end{pmatrix}, \quad (12)$$

chosen so one constraint binds at the optimum and the other does not, giving complementary slackness something visible to say.

L is strictly convex in x for every λ (its Hessian is $P \succ 0$), so the infimum in (3) is attained at the stationary point:

$$Px + q + A^T \lambda = 0 \implies x(\lambda) = -P^{-1}(q + A^T \lambda), \quad (13)$$

$$g(\lambda) = -\frac{1}{2}(q + A^T \lambda)^T P^{-1}(q + A^T \lambda) - b^T \lambda. \quad (14)$$

Discarding a constant, maximising g over $\lambda \geq 0$ is the **non-negatively constrained QP**

$$\min_{\lambda \geq 0} \frac{1}{2} \lambda^T H \lambda + d^T \lambda, \quad H = AP^{-1}A^T \succeq 0, \quad d = b + AP^{-1}q. \quad (15)$$

The dual of a QP is a QP, but over the far simpler set $\lambda \geq 0$ – the practical payoff of duality here.

3.2 Solving the dual: exact cyclic coordinate ascent

Fixing all coordinates but the i -th, (15) restricts to the scalar parabola $\frac{1}{2}H_{ii}\lambda_i^2 + \lambda_i(d_i + \sum_{k \neq i} H_{ik}\lambda_k)$. Its minimiser over the half-line $\lambda_i \geq 0$ is the vertex, projected:

$$\lambda_i \leftarrow \max\left(0, -\left(d_i + \sum_{k \neq i} H_{ik}\lambda_k\right) / H_{ii}\right). \quad (16)$$

Each update is exact and in closed form – no step size, no line search. Sweeps run until the largest coordinate change falls below 10^{-12} . Because the feasible set is a product of half-lines, coordinate-wise minimisation cannot stall at a non-stationary point, the difficulty that makes coordinate descent unreliable for non-separable constraints. Coordinates with $H_{ii} \approx 0$ are skipped, handling degenerate constraint rows. The primal is then recovered from (13), which is the stationarity condition, so (7) holds to machine precision automatically.

3.3 Solving the primal independently: logarithmic barrier method

Recovering x^* from λ^* would make p^* and d^* two readings of one computation, so the gap would prove nothing. The primal is therefore solved a second time by an unrelated route. For $t > 0$ define

$$\psi_t(x) = t\left(\frac{1}{2}x^T Px + q^T x\right) - \sum_{i=1}^m \log(b_i - a_i^T x), \quad (17)$$

whose barrier term is finite exactly on $Ax < b$ and diverges at the boundary. Its gradient and Hessian are $\sum_i a_i/(b_i - a_i^T x)$ and $\sum_i a_i a_i^T / (b_i - a_i^T x)^2 \succeq 0$, so ψ_t is strictly convex with a unique minimiser $x^*(t)$; the curve $t \mapsto x^*(t)$ is the **central path**, and $f_0(x^*(t)) - p^* \leq m/t$.

The implementation uses the classical two loops. The outer loop starts at $t = 1$ and multiplies by $\mu = 12$ until $m/t < 10^{-10}$. The inner loop applies Newton's method, solving $\nabla^2 \psi_t \Delta x = -\nabla \psi_t$ by LU with partial pivoting and stopping on the **Newton decrement** $\lambda_N^2 = -\nabla \psi_t^T \Delta x$ when $\lambda_N^2/2 \leq 10^{-12}$; the decrement is used because it is affine-invariant and estimates $\psi_t(x) - \inf \psi_t$ directly. Backtracking ($\alpha = 0.25$, $\beta = 0.5$) accepts a step only when it is **strictly feasible** and satisfies the Armijo condition, which is what keeps iterates in the domain of the logarithms.

A strictly feasible start is produced by minimising the smooth squared-hinge measure $\Psi(x) = \sum_i \max(0, a_i^T x - b_i + \delta)^2$ with $\delta = 10^{-3}$, by gradient descent with an adaptive step. Squaring removes the kink of $\max(0, \cdot)$, and $\Psi(x) = 0$ certifies strict feasibility with margin δ – so the same routine both initialises the barrier method and answers Slater's condition constructively.

3.4 Results, complementary slackness, and sensitivity

The two solvers return

$$x^* = (0.5, 1.5)^T, \quad \lambda^* = (0.5, 0)^T, \quad p^* = d^* = -4.25, \quad \text{gap} = 0. \quad (18)$$

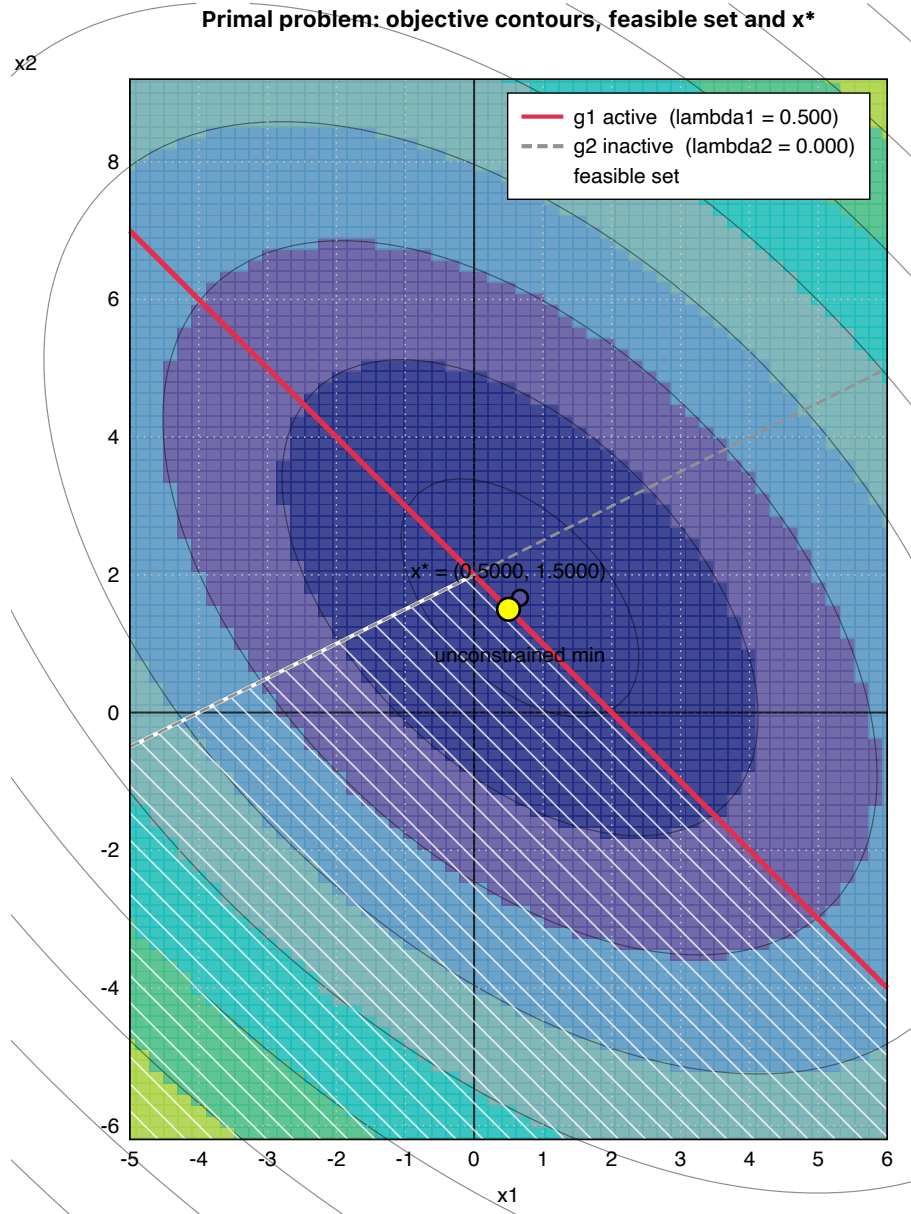


Figure 1: Instance A, primal geometry. Filled bands are level sets of f_0 , the hatched wedge is $\{x : Ax \leq b\}$, the active constraint is solid red and the inactive one dashed.

The barrier method, which never touches the dual, returns -4.249999999984 — agreement to 1.6×10^{-11} , which is what makes the zero gap meaningful. Complementary slackness is exact: constraint 1 binds ($\lambda_1^* = 0.5$, slack 0), constraint 2 is slack ($\lambda_2^* = 0$, slack 1.5), and both products vanish, exactly as (6) requires.

Figure 1 shows the geometry: objective contours, the feasible polyhedron, the unconstrained minimiser $-P^{-1}q = (2/3, 5/3)$ lying outside it, and x^* where the innermost reachable contour touches the active constraint — the geometric content of (7). Figure 2 plots g over the non-negative quadrant: a single concave hill whose summit lies **on the boundary** $\lambda_2 = 0$, the dual-side image of constraint 2 being inactive.

Figure 3 perturbs the first constraint to $a_1^T x \leq b_1 + u$, re-solving at each u , and compares with the tangent of Theorem 2. The numerical derivatives match the multipliers: $\partial p^* / \partial u_1 = -0.5 = -\lambda_1^*$ to 5.6×10^{-12} , and $\partial p^* / \partial u_2 = 0 = -\lambda_2^*$. The curve is equally instructive where it **stops** matching: for $u > 0$ it flattens, because relaxing an already-relaxed constraint eventually makes it inactive. That is the active set changing, and it is exactly where p loses differentiability and the tangent only bounds, as Theorem 2 says it must.

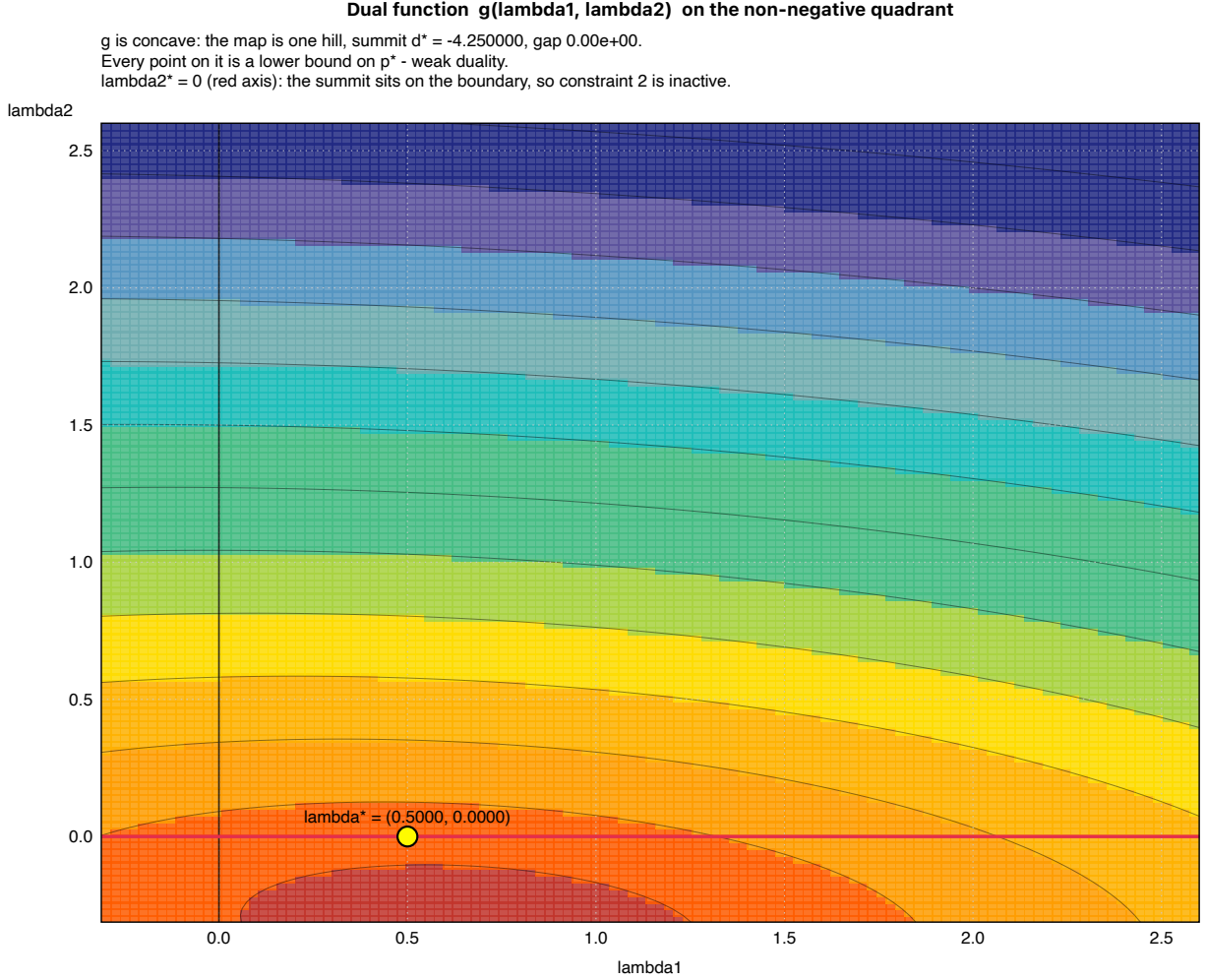


Figure 2: Instance A, the dual function $g(\lambda_1, \lambda_2)$ on $\lambda \geq 0$. Warm colours are high. Every point is a lower bound on p^* (Theorem 1); the summit attains it.

4 Instance B: resource allocation and water filling

4.1 Model, separable dual, and the water-filling rule

$$\max_{x \geq 0} \sum_{i=1}^n w_i \log(1 + x_i/\alpha_i) \quad \text{subject to} \quad \sum_{i=1}^n x_i \leq B, \quad (19)$$

with $w_i, \alpha_i > 0$. As a minimisation, $f_0(x) = -\sum_i w_i \log(1 + x_i/\alpha_i)$, which is convex, subject to $\sum_i x_i - B \leq 0$.

Only the budget is dualised; the constraints $x \geq 0$ are kept in the domain of the inner minimisation. This leaves one dual variable, so the dual function is a curve that can simply be drawn – and it is what makes the closed form below come out. The Lagrangian then separates over i , and each scalar piece is convex with derivative $-w_i/(\alpha_i + x_i) + \lambda$, giving

$$x_i(\lambda) = \max(0, w_i/\lambda - \alpha_i). \quad (20)$$

This is the classical **water-filling** rule: w_i/λ is a **water level**, α_i the **floor** of basin i , and x_i the depth of water above that floor. Basins whose floor already lies above their level receive nothing. For $\lambda \leq 0$ the Lagrangian is unbounded below, so the dual domain is $\lambda > 0$.

Each $x_i(\lambda)$ is non-increasing in λ , so the total $T(\lambda) = \sum_i x_i(\lambda)$ is continuous and strictly decreasing where positive, with $T \rightarrow \infty$ as $\lambda \rightarrow 0^+$ and $T \rightarrow 0$ as $\lambda \rightarrow \infty$. Hence $T(\lambda) = B$ has a **unique** root, located by bisection: the bracket is found by doubling, then 300 halvings. Bisection is the right tool despite its linear rate

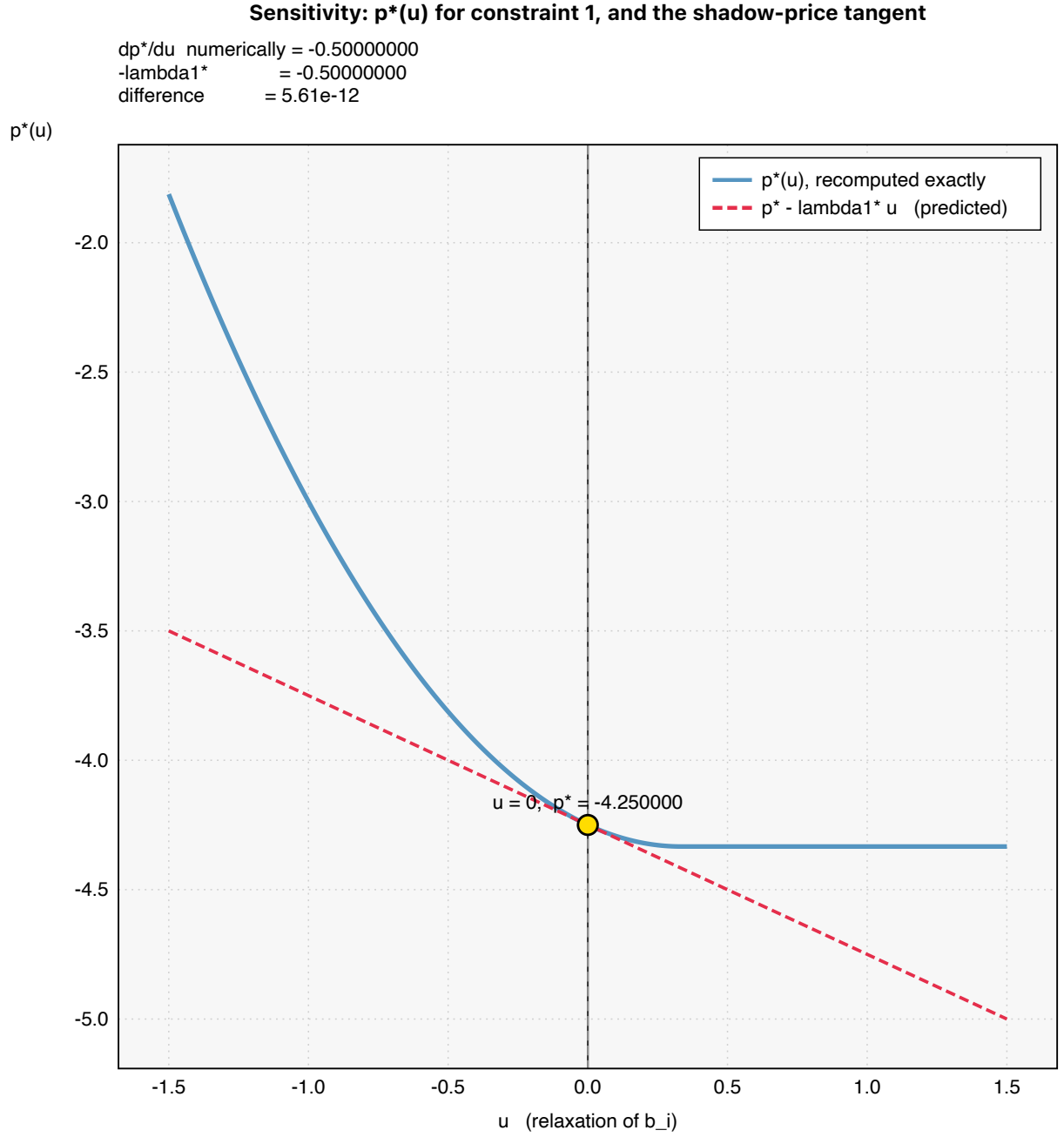


Figure 3: Instance A, sensitivity. Solid: $p^*(u)$ recomputed exactly. Dashed: the prediction $p^* - \lambda_1^* u$. Tangency at the origin is (9).

— T has kinks wherever a basin enters or leaves the water, which would need care with Newton's method, whereas bisection needs only monotonicity and continuity.

At the root the budget binds, and the KKT conditions read

$$w_i/(\alpha_i + x_i^*) = \lambda^* \text{ if } x_i^* > 0, \quad w_i/\alpha_i \leq \lambda^* \text{ if } x_i^* = 0, \quad (21)$$

the market-clearing reading: at the optimum all funded activities are **equally** worth funding, and anything unfunded is not worth the price of the budget even at zero allocation.

4.2 Results

With $w = (3, 2, 1, 0.5)$, $\alpha = (1, 0.5, 2, 0.2)$, $B = 6$:

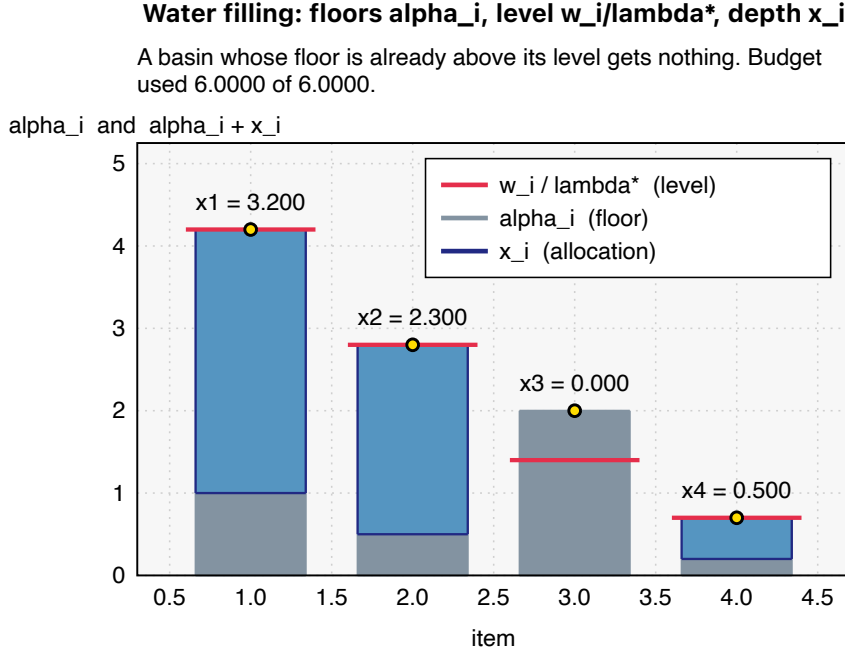


Figure 4: Instance B. Grey bars are floors α_i , blue the allocation x_i , red segments the levels w_i/λ^* . Basin 3's floor is above its level, so it stays dry.

i	w_i	α_i	x_i^*	marginal utility $w_i/(\alpha_i + x_i^*)$
1	3.0	1.0	3.2	0.714285714
2	2.0	0.5	2.3	0.714285714
3	1.0	2.0	0.0	0.500000000 — shut out
4	0.5	0.2	0.5	0.714285714

with $\lambda^* = 0.714285714$, utility 8.3771683 and $p^* = d^*$ to twelve digits. The table is (21) made concrete: the three funded activities share the common marginal utility λ^* , while activity 3 — whose floor is high relative to its weight — sits at 0.5, below the price, and is correctly shut out. The shadow price is confirmed numerically, $\partial p^*/\partial B = -\lambda^*$ to 2.4×10^{-11} : one extra unit of budget buys λ^* extra utility.

5 Instance C: a non-convex problem with a duality gap

5.1 Model and method

$$\min_{x \in \mathbb{R}} f_0(x) = x^4 + c_2 x^2 + c_1 x \quad \text{subject to} \quad x \geq x_{\min}, \quad (22)$$

with $c_2 = -6$, $c_1 = 1$, $x_{\min} = 0$. For $c_2 < 0$ this is a **double well**; with these defaults the global minimum lies at $x \approx -1.7589$ with value ≈ -10.7524 , and **the constraint excludes it**. Writing the constraint as $f_1(x) = x_{\min} - x \leq 0$,

$$g(\lambda) = \lambda x_{\min} + \inf_x [x^4 + c_2 x^2 + (c_1 - \lambda)x], \quad (23)$$

so **dualising merely shifts the linear coefficient**, $c_1 \mapsto c_1 - \lambda$, and one routine evaluates both f_0 and g . Since f_0 is coercive the infimum is finite for every λ , so g is real-valued — and concave by Proposition 1, **even though f_0 is not convex**.

Both f_0 and g need **global** minimisation, so the implementation uses a 4001-point grid scan followed by 120 golden-section iterations within the two cells adjacent to the best grid point. For a coercive quartic with at most three critical points a grid this fine cannot miss the global basin, and the refined bracket is unimodal by construction. The outer maximisation over λ uses the same scheme, legitimately, because g is concave. The

convex envelope of the sampled $p(u)$ is computed as a **lower convex hull** by a monotone-chain sweep in $O(N)$, popping a point whenever it lies on or above the chord joining its neighbours.

5.2 Results: the gap, three ways

$$x^* = 1.6887758, \quad p^* = -7.2893095, \quad \lambda^* = 1.0, \quad d^* = -9.0, \quad \text{gap} = 1.7106905. \quad (24)$$

$d^* = -9$ is exact and verifiable by hand: at $\lambda = 1$ the shifted linear coefficient vanishes, so (23) reduces to $\inf_x (x^4 - 6x^2) = -9$ at $x = \pm\sqrt{3}$. That the maximum falls there is the subgradient condition $-g'(\lambda) = x_{\min} - x(\lambda)$, and at $\lambda = 1$ the inner problem has **two** minimisers, so the subdifferential is $[-\sqrt{3}, \sqrt{3}] \ni 0$. The kink at the summit of Figure 7 is exactly this non-uniqueness.

Route	Value at $u = 0$
Primal optimum $p^* = p(0)$	-7.2893094547
Dual optimum $d^* = \max_{\lambda \geq 0} g(\lambda)$	-9.0000000000
Convex envelope ($\text{conv } p$)(0), from the hull	-8.9992382400

The envelope value matches d^* to 7.6×10^{-4} , the grid resolution — a direct numerical confirmation of (10).

The instance separates the two theorems cleanly. **Weak duality survives** — $d^* \leq p^*$ still holds and g still certifies a lower bound, its proof having used only sign conditions. **Strong duality fails** — strictly feasible points exist in abundance here (every $x > 0$), so it is precisely the convexity hypothesis of Slater's condition that has been removed, and the conclusion fails with it. The practical lesson is the standard one: in non-convex optimisation the dual is still worth solving as a source of certified bounds, and (10) says in advance how weak those bounds can be.

6 Implementation

6.1 Architecture

The application is about 4100 lines of C++20 on the natID framework, split so that the mathematics is independent of the interface.

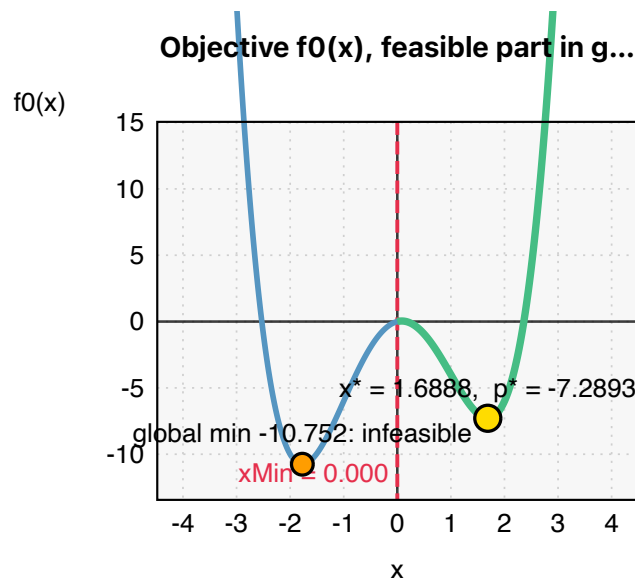


Figure 5: The objective: the global minimum at $x \approx -1.76$ is infeasible, and the constrained optimum sits in the right-hand well.

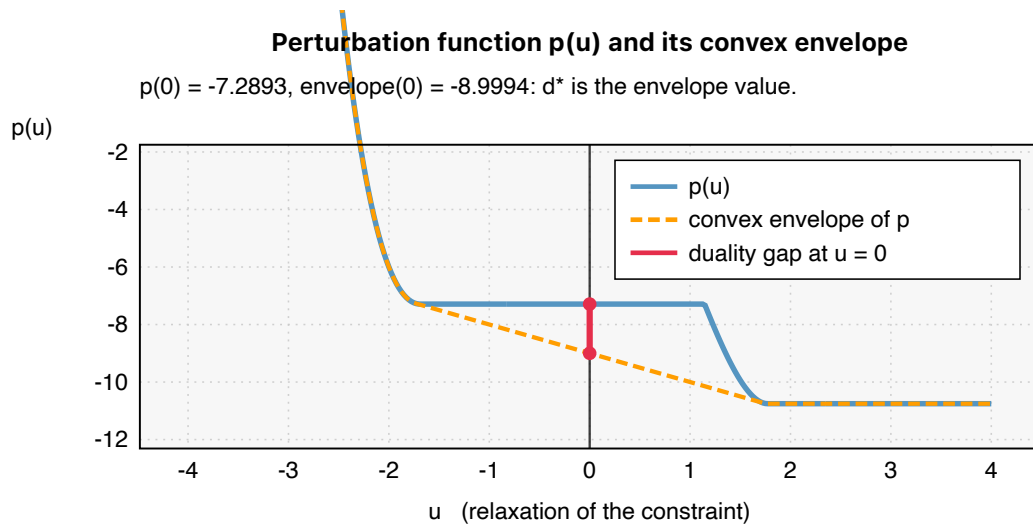


Figure 6: $p(u)$ and its convex envelope. p is flat at -7.289 , then drops to -10.75 near $u \approx 1.7$ when the relaxation finally admits the deeper well. That shape is not convex; the envelope cuts the chord across it, and the red segment at $u = 0$ is the gap.

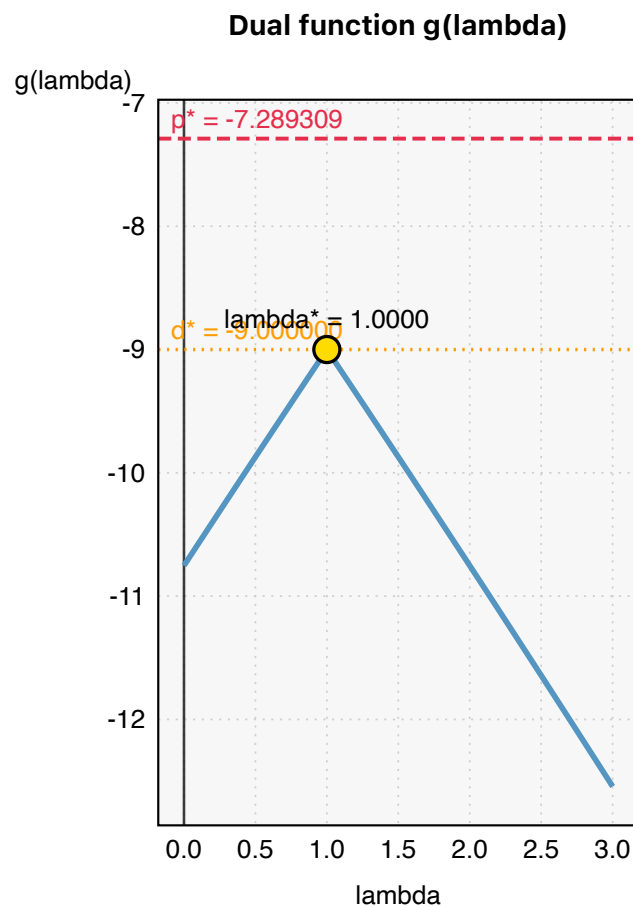


Figure 7: The dual function: concave despite the non-convex primal, and staying strictly below p^* .

File	Lines	Responsibility
core/LinAlg.h	219	dense LU, Cholesky, vector primitives
core/QPProblem.h	402	dual coordinate ascent, barrier primal, phase I
core/AllocationProblem.h	147	water filling, bisection
core/NonconvexProblem.h	217	grid + golden section, convex envelope
core/AppModel.h	141	the three instances, half-plane clipping
views/PlotCanvas.h	821	plotting engine
views/View*.h	1550	six tabs

The `core/` directory has **no GUI dependency whatsoever**. This was deliberate and had a practical payoff: the numerics were developed and verified by a standalone console program compiled against `core/` alone, before any drawing code existed, and every number in Section 7 comes from that harness. A single `AppModel` owns the three instances; every parameter panel calls one shared callback that re-solves and repaints all six tabs, so the application cannot display mutually inconsistent state.

6.2 Numerical core

Cholesky $P = LL^T$ is used wherever P^{-1} appears. Beyond the usual factor-of-two saving it is **backward stable without pivoting** for symmetric positive definite matrices, and its failure is **diagnostic**: a non-positive radicand means P is not positive definite, so the same routine doubles as the positive-definiteness test reported in the interface. **LU with partial pivoting** solves the Newton systems, where near the boundary the barrier Hessian has entries of order 10^{10} and pivoting guards against the loss of positive definiteness that rounding can induce. Forming $H = AP^{-1}A^T$ never inverts P : each row a_i is passed through the triangular solves, giving m solves instead of an inversion, both cheaper and better conditioned.

```
static QPSolution solve(const QPData& qp)
{
    sol.spd = la::isSPD(qp.P);           // Cholesky, doubles as the PD test
    if (!sol.spd) return sol;           // dual is -infinity everywhere
    la::cholesky(qp.P, L);

    sol.lambda = solveDual(qp, L, sol.dualIters); // coordinate ascent, dual QP
    xOfLambda(qp, L, sol.lambda, sol.x);        // x* = -P^-1 (q + A'lam)
    sol.pStar = f0(qp, sol.x);
    sol.dStar = dualValue(qp, L, sol.lambda);
    sol.gap = sol.pStar - sol.dStar;

    sol.slater = findStrictlyFeasible(qp, x0); // phase I, also answers Slater
    sol.barrierOk = solvePrimalBarrier(qp, sol.xBarrier, sol.pStarBarrier);
    return sol;
}
```

For the default data the coordinate ascent converges in a **single sweep**: the first update sets λ_1 exactly, the second finds λ_2 's unconstrained optimum negative and clamps it to zero, and a second sweep reproduces both.

A bug worth recording. An early `maxViolation` computed $\max_i (a_i^T x - b_i)$ but initialised the running maximum to 0 instead of $-\infty$, so it could never return a negative value and **strict** feasibility became indistinguishable from boundary feasibility. Slater's condition was reported as failing for problems that plainly satisfy it, and the barrier line search — which accepts a step only when the violation is negative — rejected every candidate. The symptom was misleading because the dual path was unaffected and still produced the correct x^* ; only the independent cross-check had silently stopped running. This is an argument for cross-validation as a design principle rather than a finishing touch.

6.3 Plotting engine

`PlotCanvas` is a small plotting library written directly on the natID `gui::Canvas` primitives, because the SDK installation was missing `gui/plot/Plot.h` and `natPlot` could therefore not be compiled against; writing the renderer directly also gave the control needed for filled contour fields and envelope overlays. Data coordinates map affinely to device pixels, with an optional equal-aspect mode that expands whichever range is short — necessary in Figure 1, where perpendicularity of ∇f_0 to the contours would otherwise be visually false. Axis ticks use the 1–2–5 decade rule, and label precision is derived from the chosen step.

Scalar fields are sampled on a regular grid and drawn as filled contour bands with curves overlaid by **marching squares**, locating edge crossings by linear interpolation. Band spacing is deliberately non-uniform — quadratic for the objective field to concentrate bands near the minimum, and its complement for the dual field to concentrate them near the summit; uniform spacing wastes almost all bands on the steep outer region. Feasible sets are computed by **Sutherland–Hodgman clipping** of a bounding rectangle against each half-plane, and shaded by 45° **hatching** (scanlines in a rotated frame, clipped by Liang–Barsky). Hatching rather than a solid fill is not decoration: a filled polygon would hide the contours underneath, and their relation to the feasible set is the whole content of the figure.

6.4 Interface, export, and safeguards

Six tabs sit on a `gui::StandardTabView`; parameter panels use `gui::NumericEdit` controls bound to model fields, with a guard flag suppressing the change callback while controls are populated from the model. Every figure here was produced in batch: `-export=<folder>` starts a timer that brings one tab on screen per tick and exports the previous tab’s canvases through `gui::Canvas::exportToPDF`. Stepping on a timer rather than looping is what makes it work — a canvas must be laid out before it has a size to render into. The canvas palette is chosen from the **rendering backend** rather than the system theme: on screen it follows dark or light mode, but any other backend always uses the light palette, because a page is white. Without this, a dark-mode foreground produced white-on-white PDFs in which the plot lines were visible but every label was invisible.

Because the interface allows arbitrary edits, each solver degrades explicitly: an indefinite P stops the solve and the interface explains that the inner minimisation is unbounded below; a failed phase I skips the barrier method and reports Slater as unsatisfied while the dual still provides a valid bound; degenerate rows are skipped by the coordinate ascent; and non-finite perturbed values are skipped by the hull construction.

7 Numerical verification

All figures below come from the standalone console harness compiled against `core/` alone.

Check	Measured as	Residual
KKT stationarity (7)	$\ Px^* + q + A^T\lambda^*\ _\infty$	5.55×10^{-16}
Complementary slackness (6)	$\max_i \lambda_i^*(b_i - a_i^T x^*) $	0.0
Primal feasibility	$\max_i (a_i^T x^* - b_i)_+$	0.0
Dual feasibility	$\max_i (-\lambda_i^*)_+$	0.0
Two independent primal solves, A	dual route vs. log barrier	1.6×10^{-11}
Shadow price, A constraint 1	central difference vs. $-\lambda_1^*$	5.6×10^{-12}
Shadow price, A constraint 2	central difference vs. $-\lambda_2^*$	0.0
Shadow price, B budget	central difference vs. $-\lambda^*$	2.4×10^{-11}
Convex envelope, C	hull at $u = 0$ vs. d^*	7.6×10^{-4}

Instance	p^*	d^*	gap	strong duality
A — convex QP	−4.250000	−4.250000	0.0	holds
B — allocation	−8.377168	−8.377168	0.0	holds
C — non-convex	−7.289309	−9.000000	1.710691	fails

Weak duality $d^* \leq p^*$ holds in all three rows, as Theorem 1 requires without exception. Strong duality holds exactly where Slater’s hypotheses are met, and fails where convexity was removed.

8 Conclusions

The dual function was derived in closed form for the QP (14) and the allocation problem (20), and by global minimisation for the non-convex instance (23); in all three it came out concave, as Proposition 1 requires. Weak duality held everywhere, consistent with a proof that never invokes convexity. Strong duality held exactly where Slater’s hypotheses were satisfied, and the non-convex instance — which has strictly feasible points but a non-convex objective — produced a gap of 1.7107, isolating convexity as the hypothesis doing the work. The identity $\partial p^* / \partial u_i = -\lambda_i^*$ was confirmed to 10^{-11} or better in all three tested cases, including the degenerate one where an inactive constraint correctly priced at zero, and the identity $d^* = (\text{conv } p)(0)$ was verified directly against the computed lower convex hull.

Two methodological points stand apart from the mathematics. Solving each convex primal **twice by unrelated methods** turned the duality gap from a tautology into a test — and that test caught a real defect that the dual path alone would have concealed. Keeping the numerical core free of any interface dependency made it testable in isolation, which is where every result in Section 7 was produced.

Possible extensions. Equality constraints with free-sign multipliers; dualising the non-negativity constraints in Instance B to obtain the full $(n + 1)$ -dimensional dual; an active-set or interior-point QP solver for larger m ; and a search on the constraint level to locate exactly where the active set changes in the sensitivity curve.

9 References

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- [4] D. P. Bertsekas, *Nonlinear Programming*, 3rd ed., Athena Scientific, 2016 — duality gaps in non-convex problems.
- [5] natID framework documentation and SDK examples, Faculty of Electrical Engineering, University of Sarajevo.